

# 11 Immunity

## AS · 2026 examination · Concept guide and original exam-style practice

### Understand the concept

Phagocytes engulf pathogens non-specifically; lymphocytes respond to specific antigens. B cells differentiate into plasma cells producing antibodies and memory cells enabling faster secondary responses. T helper cells coordinate responses; T killer cells destroy infected cells. A vaccine supplies antigen safely to induce immune memory; it does not directly deliver permanent ready-made antibodies.

### Worked example: follow the reasoning

On a second exposure to the same antigen, memory cells divide faster and produce a larger antibody response before symptoms develop.

### Quick check · multiple choice

A What enables a faster secondary antibody response? [1]

- A Erythrocytes
- B Platelets
- C Memory cells
- D Antigens disappear forever

Write your choice: \_\_\_\_\_

### Structured practice

Answer in the space provided. Show working and units where relevant.

1 Distinguish active vaccination from passive antibody transfer in onset and duration of protection. [3]

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## Answers and explanation

Quick check: C. Memory cells

### Question 1 [3 marks]

Passive transfer supplies antibodies and acts immediately [1], but is temporary as antibodies degrade [1]; active vaccination induces memory cells and lasts longer after a slower primary response [1].

#### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.